



Assessment 3
6 June
2007

Mathematics

General Instructions

- Reading Time – 2 minutes
- Working Time – 1 hour 10 min
- Write using black or blue pen
- Board approved calculators may be used
- A table of standard integrals is provided at the back of this page
- All necessary working should be shown in every question

Total Marks – 50

Attempt Questions 1–5
All questions are of equal value

Begin each question on a NEW PAGE.

NAME: _____ TEACHER: _____

Question	H3	H5	H6	H8	Total
1	/1	/9			/10
2	/3	/2		/5	/10
3		/3		/7	/10
4	/2	/4		/4	/10
5			/10		/10
TOTAL	/6	/18	/10	/16	/50

Question 1 (10 marks)**Marks**(a) Differentiate with respect to x

(i) $y = (e^x + 4)^7$

2

(ii) $f(x) = \ln(x^2 + 1)$

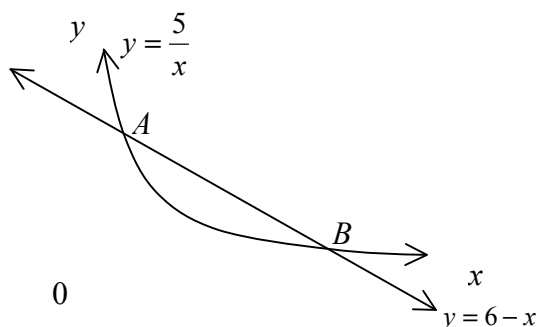
1

(b) Find

(i) $\int \sqrt{x} dx$

1

(ii) $\int (4x + 2)^5 dx$

2(c) Evaluate $\int_{-1}^2 (5 + e^{3x}) dx$ correct to 1 decimal place.**3**(d) Use the change of base formula to evaluate $\log_5 9$ correct to 3 decimal places.**1****Question 2 (10 marks) Start a new page.****2**(a) Differentiate $\log_e \frac{x+7}{x-3}$ **3**(b) Solve $2 \log_e x = \log_e (6 - 5x)$.(c) The line $y = 6 - x$ meets the curve $y = \frac{5}{x}$ at points A and B as shown in the diagram.(i) Find the co-ordinates of A and B .**2**

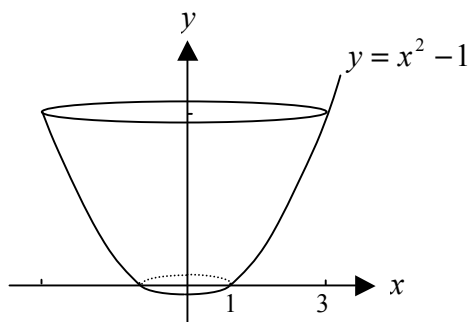
(ii) Find the shaded area between the curves.

3

Question 3 (10 marks) Start a new page.

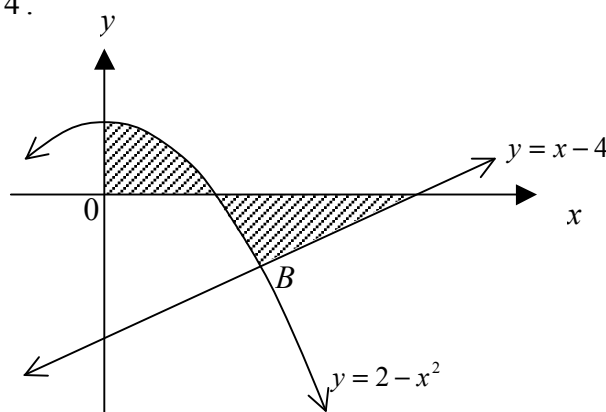
Marks
Continued overleaf

- (a) Evaluate $\int_{-2}^1 (x^3 - 4) dx$. 3
- (b) (i) Use the trapezoidal rule with 4 function values to find an approximation to the area under the curve $y = \ln x$ between $x = e$ and $x = 4e$. 2
- (ii) Is the approximation found in (i) an overestimate or an underestimate of the exact area? Justify your answer. 1
- (c) A bowl is designed by rotating the portion of the curve $y = x^2 - 1$ between $x = 1$ and $x = 3$ around the y -axis as illustrated in the diagram below. Find the volume of the solid formed. 4



Question 4 (10 marks) Start a new page.

- (a) If $\log_a p = 1.18$ and $\log_a q = 1.54$ find the value of $\log_a (apq)$ 2
- (b) (i) Differentiate $\sqrt{2x^2 + 3}$ with respect to x . 2
- (ii) Hence find $\int \frac{8x}{\sqrt{2x^2 + 3}} dx$. 2
- (c) In the diagram the shaded region is bounded by the parabola $y = 2 - x^2$, the axes and the line $y = x - 4$. 1



- (i) Show that the co-ordinates of the point of intersection B are $(2, -2)$. 1
- (ii) Write an expression for the shaded area. Do NOT find the area.

Question 5 (10 marks) Start a new page.Continued overleaf **Marks**

The function $f(x) = xe^{-2x} + 3$ has a first derivative $f'(x) = e^{-2x} - 2xe^{-2x}$ and second derivative $f''(x) = 4xe^{-2x} - 4e^{-2x}$.

- | | | |
|-----|--|---|
| (a) | Find the stationary point on $y = f(x)$. | 2 |
| (b) | For what values of x is the function increasing? | 1 |
| (c) | Show that the co-ordinates of the point of inflexion are $\left(1, 3 + \frac{1}{e^2}\right)$. | 2 |
| (d) | Determine when the curve is concave up. | 1 |
| (e) | Sketch $y = f(x)$ for $-\frac{1}{2} \leq x \leq 2$. | 3 |
| (f) | What value does $f(x)$ approach as x gets very large? | 1 |

End of paper

SOLUTIONS - 2007 MATHEMATICS Task 3 - YR 12

Q1a) $y = (e^x + 4)^7$

$\frac{dy}{dx} = 7e^x (e^x + 4)^6$ (2)

$f'(x) = \frac{2x}{x^2 + 1}$ (1)

b)(i) $\int \sqrt{x} dx = \frac{2}{3} x^{3/2} + C$
 $= \frac{2}{3} \sqrt{x^3} + C$ (1)

(ii) $\int (4x+2)^5 dx = \frac{(4x+2)^6}{24} + C$ (2)

c) $\int_{-1}^2 (5 + e^{3x}) dx = \left[5x + \frac{1}{3} e^{3x} \right]_{-1}^2$ (3)
 $= \left(10 + \frac{e^6}{3} \right) - \left(-5 + \frac{e^{-3}}{3} \right)$
 $= 14.95$

d) $\log_5 9 = \frac{\log 9}{\log 5} = 1.365$ (1)

Question 2

a) $\frac{d}{dx} \left(\log_e \frac{x+7}{x-3} \right) = \frac{d}{dx} (\log_e(x+7) - \log_e(x-3))$
 $= \frac{1}{x+7} - \frac{1}{x-3}$ (2)

b) $2 \log_e x = \log_e (6-5x)$
 $x^2 = 6-5x$ where $x > 0$ and
 $x^2 + 5x - 6 = 0$ $6-5x > 0$
 $(x+6)(x-1) = 0$ $\therefore 0 < x < \frac{6}{5}$
 $\therefore x = 1$ or -6 (3)

$\therefore x = 1$ is the only solution

c)(i) $\frac{5}{x} = 6-x$ at A and B
 $x^2 - 6x + 5 = 0$
 $(x-5)(x-1) = 0$
 $x = 1, 5$ at A $x = 1, y = 5$
 at B $x = 5, y = 1$
 $\therefore A = (1, 5)$ and $B = (5, 1)$ (2)

(ii) Area = $\int_1^5 6-x - \frac{5}{x} dx$
 $= \left[6x - \frac{x^2}{2} - 5 \log_e x \right]_1^5$
 $= 12 - 5 \ln 5$
 $= 3.95$ (2 d.p.)

Question 3

a) $\int_{-2}^1 (x^3 - 4) dx = \left[\frac{x^4}{4} - 4x \right]_{-2}^1$
 $= \left(\frac{1}{4} - 4 \right) - (-4 + 8)$
 $= -15 \frac{3}{4}$

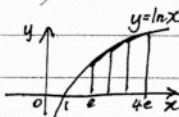
b)(i)

x	e	2e	3e	4e
ln x	ln e	ln 2e	ln 3e	ln 4e
=	1	1+ln 2	1+ln 3	1+ln 4

 Calculator:
 $\ln(2e)$

Area $\div \frac{e}{2} (1 + 1 + \ln 2) + \frac{e}{2} (1 + \ln 2 + 1 + \ln 3) + \frac{e}{2} (1 + \ln 3 + 1 + \ln 4)$
 $= \frac{e}{2} (6 + 2 \ln 2 + 2 \ln 3 + \ln 4)$ (exact)
 $= 14.91$ (2 d.p.)

(ii) The curve is concave down and the trapezia are under the curve at all times. $\therefore$ This estimate underestimates the true area.



c) $V = \pi \int x^2 dy$ $x=1, y=0$
 $= \pi \int_0^8 (y+1) dy$ $x=3, y=8$
 $= \pi \left[\frac{y^2}{2} + y \right]_0^8$ $y = x^2 - 1$
 $= \pi \left[\frac{64}{2} + 8 - 0 \right]$ $x^2 = y + 1$
 $= 40\pi$ units³

Question 4

a) $\log(apq) = \log_a a + \log_a p + \log_a q$
 $= 1 + 1.18 + 1.54$
 $= 3.72$

b)(i) $f'(x) = \frac{1}{2} (2x^2 + 3)^{-1/2} \cdot 4x$
 $= \frac{2x}{\sqrt{2x^2 + 3}}$

(ii) $\int \frac{8x}{\sqrt{2x^2 + 3}} = 4 \cdot \int \frac{2x}{\sqrt{2x^2 + 3}} dx$
 $= 4 \sqrt{2x^2 + 3} + C$

Question 4 (cont)

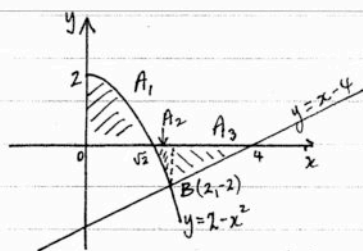
c) (i) At $x=2$ $y=x-4$ and $y=2-x^2$
 $= -2$ $= 2-2^2$
 $= -2$

$\therefore B$ has coordinates $(2, -2)$

(ii) Area = $\int_0^{\sqrt{2}} (2-x^2) dx - \int_{\sqrt{2}}^2 (2-x^2) dx - \int_2^4 (x-4) dx$

OR

Area = $\int_0^{\sqrt{2}} (2-x^2) dx + \left| \int_{\sqrt{2}}^2 (2-x^2) dx \right| + \left| \int_2^4 (x-4) dx \right|$
 or area of $\Delta = \frac{1}{2} \times 2 \times 2 \uparrow$



A_2 and A_3 are below x axis
 $\therefore \int$ will be negative for each

Question 5

a) Stat pt if $f'(x)=0$

$\therefore e^{-2x}(1-2x)=0$

As $e^{-2x} > 0$ for all x , then $x = \frac{1}{2}$ is only solⁿ

then $y = 3 + \frac{1}{2e}$

$\therefore$ Stat pt at $(\frac{1}{2}, 3 + \frac{1}{2e})$

b) Increasing if $f'(x) > 0$

$e^{-2x}(1-2x) > 0$

$\therefore 1-2x > 0$ as $e^{-2x} > 0$

$\therefore x < \frac{1}{2}$

c) $f''(x)=0$ for pt of inflexion

$4e^{-2x}(x-1)=0$

$\therefore x=1$ as $e^{-2x} > 0$

then $y = e^{-2} + 3 = 3 + \frac{1}{e^2}$

Check concavity

x	$\frac{1}{2}$	1	$\frac{3}{2}$
$f''(x)$	-	0	+

$\therefore$ Concavity changes

$\therefore$ Inflexion at $(1, 3 + \frac{1}{e^2})$

d) Concave up if $f''(x) > 0$

ie $4e^{-2x}(x-1) > 0$

$\therefore x > 1$ since $4e^{-2x} > 0$

f) As $x \rightarrow \infty$, $e^{-2x} \rightarrow 0$

$\therefore f(x) \rightarrow 3$

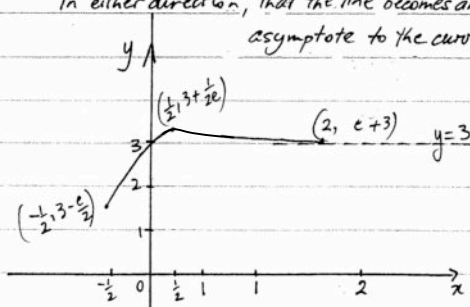
ie $f(x)$ approaches 3

$$\begin{aligned} f(x) &= xe^{-x} + 3 \\ f'(x) &= e^{-x} - 2xe^{-2x} \\ f''(x) &= 4xe^{-2x} - 4e^{-x} \end{aligned}$$

Note: The curve exists at $x=0$ and

so there is NOT an open circle at $(0, 3)$

A curve may pass through an horizontal asymptote. It is only as becomes very large in either direction, that the line becomes an asymptote to the curve



c) When $x=0$, $f(x)=3$

$x = -\frac{1}{2}$ $f(x) = 3 - \frac{1}{2e} \div 1.64$

$x = \frac{1}{2}$ $f(x) = 3 + \frac{1}{2e} \div 3.18$

$x = 1$ $f(x) = 3 + \frac{1}{e^2} \div 3.14$

$x = 2$ $f(x) = 3 + \frac{1}{e^4} \div 3.037$